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Early Warning System for Player Recovery? A Series of Case Studies Illustrating the Application of Individualised Adaptive Reference Ranges in the Longitudinal Blood Monitoring of English Premier League Soccer Players.

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Abstract

Blood biomarkers can provide objective insight into a player's physiological state of recovery. In the context of elite football, the utilisation of a point-of-care (POC) test for measuring inflammation via capillary sampling, facilitates frequent data acquisition in order to assess recovery status within game-to-game micro-cycles. The smaller within-subject variability compared to the variability between subjects limits the practical use of population-based reference ranges. Individualised approaches to biomarker monitoring may therefore be of higher potential value in assessing player health and recovery. We aimed to explore the application of individualised adaptive reference ranges (IARR) in English Premier League (EPL) soccer players using a POC biomarker for C-Reactive Protein (CRP) as a marker of inflammation. Using historical data collected from players' CRP values during the 2019-2020 season, we evaluated the effectiveness of static and IARR in identifying abnormal values and reported sensitivity and specificity at a 5% significance level. Our analysis confirmed that individualised monitoring with personalised adaptive reference ranges is more effective in identifying true abnormalities compared to PBRR (87.8% vs 86.2% for pre-match CRPs and 85.2% vs 81.5% for post-match CRPs), particularly when the intra-individual variability is considerably lower than inter-individual variability. The application of IARR for blood monitoring data could assist the practitioner in identifying periods where a player may require performance (e.g. workload management and recovery practices) or medical support from the multi-disciplinary team. However, IARR serve more as an early warning system than a diagnostic tool. Thus, significant care is needed to prevent misuse and misinterpretation when implementing this strategy in real-world settings.

Key Words: Soccer; Recovery; Inflammation; Personalised; Biomarkers; English-Premier-League

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Introduction

The high intensity demands of elite soccer impose distinctive physiological and mechanical loading demands on players.[1] The performance of concentric (e.g. accelerations/maximal sprinting), isometric (e.g. shielding the ball against offensive pressure) and particularly eccentric (e.g. jumping, landing and rapid deceleration movements) muscle contractions during the high velocity and technical actions of the game results in significant mechanical loading, leading to exercise induced muscle damage (*EIMD*).[1,2] *EIMD* can be briefly summarized into two phases; the initial primary damage that occurs as a consequence of the mechanical work performed, and a secondary inflammatory response,[3] characterized by the accumulation and infiltration of immune cells into the damaged tissue.[3-5] Evidence suggests there are substantial elevations in inflammatory markers during the first 72 hours post-match.[6-8] However, the magnitude, time course and resolution of this inflammatory response will depend on the intensity and duration of the damaging exercise and the players' susceptibility to the initial damaging stimulus.[3-5] Acute inflammation is a vital protective process, acting to promote repair and signalling for adaptation, however, overwhelming inflammation can result in secondary damage and promote maladaptive tissue remodelling,[5] or at least prolonged recovery requirements. Indeed, the rate of deterioration and recovery of muscle function and soreness post-exercise is largely dependent on the magnitude of muscle damage and inflammation.[4] A critical balance clearly exists between the beneficial and detrimental effects of inflammation, paralleling the theory of hormesis.[9] Although elite players are well adapted to their sport, the long season with consistent exposure to acute inflammatory episodes and limited periods of rest, may lead to a loss of inflammatory control and the development of chronic low-grade inflammation[10] and places the soccer player at a higher risk of under recovery and its negative effects – fatigue, under-performance, and overuse injury.[11]

Monitoring inflammation in the professional soccer player may be an important medical and performance objective for protecting player health, recovery and performance. Although, traditionally utilized for diagnostic or anti-doping purposes, blood biomarkers may provide an objective means of assessing recovery status.[12,13] The measurement of blood biomarkers of inflammation may be used to make inferences about the players underlying physiology and could provide the high-performance practitioner with beneficial information to promote recovery in the interval between successive matches. Data describing how inflammation fluctuates in response to match play and throughout a season in the English Premier League (*EPL*) are rare.[14] Indeed, most studies investigating inflammatory disturbances induced by a competitive game have done so in response to a single match,[6-8] which does not account for the recognised large match-to-match variation in physical demands,[15] or the cumulative effects of prior exercise exposure.[16] The logistical challenges and the delayed retrospective analysis associated with venous blood sampling may explain why longitudinal studies examining inflammation in players are generally absent from the literature. The utilisation of point of care (*POC*) tests for biomarkers of inflammation via capillary sampling, would allow for frequent testing in order to assess ongoing adaptation/recovery from disturbed homeostasis, and may be used to enhance the load management of players.[12]

Lewis et al.[17] outline that the successful application of biomarker monitoring programs in elite sport is not only hindered by the need for the generation of "real time" results but also an understanding of what is 'normal' and what constitutes a meaningful deviation from normal for each individual athlete. In clinical settings, population-based reference ranges, also known as normative or static reference ranges, which are generated using a cross-sectional sample of the general population, are typically used when interpreting a biomarker

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result for an individual by classifying their value as typical or atypical. Beyond a medical diagnostic or anti-doping purpose, these ranges can be of limited value for the purpose of assessing recovery status since athletes are often outliers due to their training status[18,19] and global training stress.[12] Furthermore, although these ranges may be valuable in assessing a players' health when there is only one observation recorded, they have limited applicability when there are serial measurements since consideration of changes in the biomarker over time are ignored.[20] This is particularly true when biomarkers present higher between-subject biological variation than within-subject biological variation.[20] As an example, Heikura et al.[21] recently reported that bone injuries were 4.5 fold more prevalent in male athletes with testosterone values in the lower quartile of the sample, but notably all values were within the normative reference range. Individualised approaches to biomarker monitoring may therefore be of higher potential value in protecting player health or detecting an under-recovered state. Indeed, the combination of the intra-individual variation with inter-individual variation in deriving reference ranges from repeated measurements in a given player may lead to a significant improvement of the biomarker signal. This individualisation is used in the Athlete Biological Passport, as implemented by WADA[22] and a recent report has demonstrated its potential utility for the purpose of individualizing athlete recovery in a group of elite junior swimmers and triathletes.[13]

Literature exploring the application of individualised adaptive reference ranges (*IARR*) in elite sport are rare. To the best of our knowledge, there are no published longitudinal data in elite soccer. The current study aimed to explore the application of IARR in EPL soccer players using a *POC* biomarker for C-Reactive Protein (*CRP*) as a marker of inflammation. *CRP* is widely used as a marker of systemic inflammation. It is an acute phase reactant, secreted by the liver in response to a variety of inflammatory cytokines, but predominantly under transcriptional control by the cytokine Interleukin (IL)-6 [7] and has been reported to be a sensitive marker of inflammation after a soccer match, showing a marked but transient rise within 24 hours post- match, before declining thereafter and returning near baseline levels at 72 hours post-match.[7] There are many factors that can alter baseline *CRP* levels including the cumulative effect of recent training and competition,[19] as well as alterations in lifestyle and wellness factors such as sleep quality,[23] illness[24] and dietary patterns.[25] The purpose of our investigation was to evaluate the utility of the adaptive ranges in detecting health/recovery concerns in the elite soccer player and differentiating between normal and abnormal profiles.

Methods

Participants

Twenty two players (age = 25.4 [3] y, height = 181.8 [7.8] cm, weight = 75.1 [6.2] kg, and body mass index (BMI) = 22.7 [1.5]) were recruited to participate from the first team of an *EPL* club. Ethical approval was obtained from the University of Galway research ethics committee and written informed consent was provided by all participants.

Design overview

We conducted a longitudinal analysis of 17 *EPL* matches during the 2019-2020 season. Over the course of the first half of the competitive season (August to December), *CRP* measurements were taken before breakfast from 22 players before and after 17 matches (440 total blood samples). The matches were interspersed with one week recovery. Subjective assessments of wellness and illness symptoms were also collected via questionnaire. Match load (volume, intensity, minutes) was recorded for each game. Testing was carried out between 9 am and 11 am, with pre-game measurements taken on Match day -1 (MD-1) and

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post-match measurements taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) or a combination thereof. Players were asked to report for testing in a fasted and hydrated state. Compliance with pre-testing instructions were verified via questionnaire. Training was prescribed by coaching staff and was deliberately not influenced by the study design or the research personnel. The players training load varied in intensity and volume depending on factors including but not limited to the following: player status (starter/bench player), time of season, upcoming games, injuries and illnesses.

Match load assessment

Games were analysed during the 2019-2020 season, using a multi-camera computerised tracking system (Second Spectrum Inc., CA, USA). Game minutes, total distance (km) and high speed running distance (THSR; defined as distance covered at >21km/h) were recorded each game via cameras positioned in each of the stadiums and were analysed using match-analysis software (Second Spectrum Inc., CA, USA) to produce a single data set of each player's activity during a match.

Perceived ratings of wellness and illness symptoms

Subjective assessments of well-being[26] were recorded prior to each blood sampling. This questionnaire contained 5 questions: Have you eaten today? (Yes/No); Have you exercised today? (Yes/No); Rate your energy levels (Very Low/Low/OK/Good/High); Rate your muscle soreness (Very Sore/Sore/Average/Not Sore/Feel Great); Have you had any of these symptoms recently?: Fever, Cold/Sore Throat, Headache, Joint/Muscle Ache, Diarrhoea/Sickness (Options: Today/In the last week/None).

Blood Sampling

Whole blood capillary samples (20 µL) were taken from the participant's earlobe, and immediately analysed at room temperature in line with the manufacturer's instructions. Inflammation was measured with an immunoturbidimetric high sensitivity *CRP* assay on the Cube-S analyzer (Eurolyser Diagnostica GmbH, Salzburg, Austria). This photometric measurement of CRP is based on an antigen-antibody reaction between antibodies to human *CRP* bound to polystyrene particles and *CRP* present in the sample. All the biochemistry is contained within the test cartridges supplied by the manufacturer. The manufacturer reports a coefficient of variation for the assay of 2.83% for whole blood with a correlation coefficient R^2 of 0.951 against a clinical laboratory gold standard. The measurement range of the analyzer for whole blood is 0.5-20.0 mg/L.

Data analysis: Generation of Individualized Adaptive Reference Ranges

The generation of *IARR* is based on a mixed-effect modelling framework utilizing an approximate Expectation-Maximization (EM) algorithm, as proposed by Roshan et. al.[20] In particular, reference ranges will be adapted with each new measurement for an individual combining prior information from the population with the individual's past measurements. For example let $[y_{i1}, y_{i2}, \dots, y_{in}]$ represent a sample of CRP measurements for a particular player over n different follow up times. A new measurement, $y_{i,new}$, will be being considered abnormal if it falls outside of the 95% quantiles of $P(y_{i,new}|data)$; where $data$ includes both the player's previous measurements and also data from other players in the study. The detailed methodology is described by Roshan et. al.[20] A clinical range of < 5.1 mg/L for soccer players has been previously defined by Meyer et al.[27] This range differs from the static range in our study, which is generated from longitudinal data collected from the squad during the testing period. Tolerance intervals[28] were used for generating static reference ranges. We employed a "leave-one-player-out" approach on the complete case data to construct both

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adaptive and static reference ranges. Specifically, when constructing reference ranges for a particular player, we re-estimated all relevant parameters, excluding the data from that player. This method simulates applying the models as if all readings in the series were new values to test. A prerequisite for the effective application of *IARR* is the inclusion of only "healthy" player data in the computation of the adaptive range. As such, when players reported the presence of any illness symptoms, injury, or very high ratings of fatigue and soreness to the research personnel or medical team at the club, these data were excluded from the generation of the reference ranges. As appropriate, players were also classified as injured and ill by the medical staff. Excluded data points were used to assess whether both the static and adaptive reference ranges could detect them as abnormal values. The static and adaptive reference ranges classify individual's *CRP* values as normal or abnormal based on whether they fall inside or outside the generated reference ranges at each time point. To evaluate the effectiveness of these reference ranges in identifying abnormal *CRP* values for a player, we reported sensitivity and specificity at a 5% significance level. In simple terms, sensitivity measures the percentage of correctly identified abnormal observations, while specificity measures the percentage of correctly identified normal observations. A series of five individual cases of longitudinal *POC* biomarker data are described. These cases were selected as they highlight scenarios where the application of *IARR* shows promise as a strategy to assess ongoing recovery from disturbed homeostasis in the professional soccer player, as well as highlighting scenarios where significant care is needed to avoid misuse or misinterpretation. The resulting *IARR* are displayed alongside static reference ranges (based on the group data only) in Figures 1–5, with the *IARR* represented by a solid blue line and the static ranges shown as a dashed line. Excluded data points are all highlighted in red and do not affect the future calculation of the adaptive ranges. The R programming language (version 4.3.1) was used for the analysis. Microsoft Powerpoint was used to add additional context to the figures.

Results

Table 1 summarises the performance of the two static and adaptive reference ranges in terms of their average sensitivity and specificity. As can be seen, although the average specificity for *IARR* is lower than for static reference ranges in both pre- and post-match *CRP* values, *IARR* outperforms static reference ranges in terms of sensitivity. This indicates that IARR is more effective in identifying true abnormal values, albeit with a slightly higher false positive rate. Notably, a closer examination of the data revealed that 15% (4 out of 27) of pre-match *CRP* and 6% (1 out of 16) of post-match *CRP* false positives identified by IARR occurred just before an actual abnormal value, suggesting that *IARR* could serve as an effective early warning system. Furthermore, 15% (4 out of 27) of pre-match *CRP* false positives and 15% (1 out of 12) of post-match *CRP* false positives were identified immediately after a true abnormal value, indicating that the players may not have fully recovered, as their *CRP* values remained outside their personalised reference ranges.

The analysis further revealed that *IARR* is a most effective method when the intra-individual variability is considerably lower than inter-individual variability. **Table 2** presents a detailed analysis of the performance of the two methods, categorized by the ratio of intra-individual variability to inter-individual variability. As can be seen, for ratios less than 0.25, *IARR* demonstrates superior sensitivity, achieving 100% for both pre- and post-match *CRP* measurements, compared to the static reference range's 50% and 83.3%, respectively. As the ratio increases to between 0.25 and 0.5, *IARR* maintains high sensitivity for post-match *CRP* compared to the static reference range. However, these heightened sensitivity in *IARR* comes

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with slightly lower specificity. This suggests that for low ratios, *IARR* is more effective at identifying outliers because of its narrower range, although it produces more false positives. In contrast, the static reference range, which is wider, fails to detect true abnormalities but does not generate any false positives. At higher ratios ([0.5, 1), [1, 2), and [2, 5)), the specificity of *IARR* improves, although sensitivity slightly declines, becoming comparable to or even less than that of the static reference range method. This is mainly because *IARR* produces reference ranges that are as wide or even wider than the static reference ranges, which enhances specificity but at the expense of sensitivity. This trend indicates that *IARR* is more effective at lower ratios but trades off some sensitivity for improved specificity at higher ratios. Notably, for ratios above 5, data is sparse, yet *IARR* and static reference range methods show comparable sensitivity and specificity for pre-match *CRP* measurements. But the scarcity of data at this level necessitates cautious interpretation of these findings.

The five players selected as case studies where the use of static and adaptive reference ranges in monitoring their *CRP* values will be reviewed in more detail are presented in **Tables 3-7** and **Figures 1-5**.

Discussion

In this study, we describe the use of personalised adaptive reference ranges to evaluate biomarker data collected longitudinally from soccer players within an EPL season. Using historical data of elite soccer players' *CRP* values, our analysis showed that individualised monitoring with personalised adaptive reference ranges is more effective in identifying true abnormalities compared to population-based static reference ranges, particularly when the intra-individual variability is considerably lower than inter-individual variability. However, it should be noted that a larger sample along with a formal statistical test is required to assess the significant differences between the two methods in terms of their sensitivity and specificity. The case series presented demonstrate that *CRP* displayed negligible heterogeneity of intra-individual variability in healthy players, and that players with chronic low-grade inflammation present markedly higher intra-individual variability, which has a significant impact on their respective individualized ranges. The application of individualised reference ranges for blood biomarker monitoring data could assist the practitioner in identifying periods where a player may require performance (e.g. workload management and recovery practices) or medical support from the multi-disciplinary team. To further support practitioners, an online web-based application has been developed, available at https://pci-nuig.shinyapps.io/barshinydemo/, allowing users to import their data and generate the required reference ranges for their specific subject of interest. However, significant care is needed to prevent misuse and misinterpretation when implementing this strategy in real-world settings. Indeed, four main themes emerged from the data for the high-performance practitioner to consider:

1. *Pre-analytical considerations: Establish Baseline values*

It is important to build a valid benchmark profile for making confident comparisons of changes in biomarker responses.[16] There is a need to establish baseline values whilst the player is in a healthy physiological state (e.g. frequent *POC* testing performed when the player returns from the off-season period), as highlighted in the case description for Player 3 (**Table 4**). This approach allows the practitioner to gather baseline values, generate normative data and observe natural oscillations in the biomarker results around a homeostatic set point during healthy training periods, establishing an *IARR* for each player. The *IARR* can continue to be modelled as new data points are generated, resulting in the construction of individual

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player profiles prior to the commencement of the season. This can also assist the practitioner in understanding a players' recovery state when they are in a return to play phase after rehabilitation from an injury, or during important monitoring windows such as when a player returns from international duty (**Table 7**), or an illness (**Figures 1-3**), as baseline data can be used as a reference point when assessing a players' response to a given training session or match. In the case of the latter, our data suggests monitoring inflammation longitudinally may prove beneficial to the elite practitioner in identifying illness prone players who could benefit from medical support (**Table 4**). The absence of baseline data collected prior to the season can therefore be considered a limitation in the current study.

2. *Timing of Testing: Distinguish between different physiological states*

We applied an *IARR* to post-match biomarker responses, as previously conducted by Hecksteden et al.[13], to aid the sport science practitioner in evaluating the players level of post-match fatigue. Indeed, testing administered pre- and post-match proved valuable in detecting recovery concerns for Player 1 (**Figure 1**). However, a limitation in our approach is that we did not apply a separate *IARR* to each post-game biomarker measure (i.e. MD+1, MD+2, MD+3), given the time-dependant changes in circulating markers of inflammation in response to match-play.[6-8] It is noteworthy that anecdotal evidence collected by the present authors suggests that the timeline of the inflammatory response is highly individual and is associated with contextual differences across matches. Nevertheless, these testing days essentially represent different physiological states and should be interpreted as such. Furthermore, opportunities for data collection are frequently result-dependent and teams often vary the scheduling of 'recovery days'. Additionally, logistical challenges with data collection may mean that players miss testing appointments as they balance their other professional commitments. This is evident by the number of missing post-match data points in our case series (**Figures 1-5**). Therefore, a flexible approach to testing seems the most appropriate. Once enough data has accumulated, players may have a separate IARR for each testing day for a typical micro-cycle.

3. *Analytical Considerations: Interpreting meaningful changes*

A challenge for sports scientists is to identify meaningful changes in biomarker data in order to make informed decisions regarding player management. Such decisions need to be made within 1 to 2 hours before training.[37] The statistical methods presented here may assist with this process, however, it is important to keep in mind that human interpretation is required in order to intuitively observe trends in the data, avoid misinterpretation of the *IARR* and make informed decisions regarding player management (**Table 4**). A practical approach is therefore necessary whereby practitioners judge abnormal data points identified by *IARR* in a live manner (i.e. at the time of testing) based on a combination of factors including statistical methods, their own domain knowledge and relevant context collected from the player, staff and other data streams, so that meaningful changes can be distinguished from natural variations in the biomarker (e.g. Player 4 pre-match 3 **Figure 4A**). The lack of this judgement or context may result in the *IARR* including data points that should not be incorporated into generation of the next adaptive reference ranges. Our analysis was limited by the willingness of the player to detail such information in the wellness questionnaire. Self-reporting is influenced by a number of factors such as match outcome[38] and social influences,[16] which may result in under-reporting of unfavourable responses to the sports scientist. Anecdotally, this is largely dependent on the individual in question, with some athletes consistently reporting the same subjective wellness throughout the study period regardless of time of measurement (**Table 6**).

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4. *Context before action: A multidisciplinary team approach*

The individualized approach to biomarker monitoring presented in this study represents a promising strategy to assess ongoing recovery from disturbed homeostasis in the professional soccer player and could serve as an early warning system for identifying abnormal profiles for a given player during the competitive season (**Figure 5**). In real-world scenarios, it is likely to be rare for blood biomarker and monitoring data to influence player match selection, however, this information may be used to better inform the subsequent course of action (e.g. prescription of recovery modalities, adjustment of training workload) to protect player health, promote timely recovery and ensure players are not overloaded in the lead up to games. The highly contextualized nature of biomarker testing and analysis means that communication amongst key stakeholders including the sports scientist, team doctor and the player themselves is vital in understanding the biomarker results. Furthermore, when blood biomarker data is combined with other data sources routinely collected in the high-performance environment such as workload data, subjective wellness, neuromuscular function tests, and data derived from wearable devices (e.g. heart rate variability, sleep actigraphy), the practitioner may be able to gather a more holistic understanding of each individual player. The ability to integrate data also allows for information to be shared across disciplines and enables practitioners to better understand how one factor affects another (e.g. training load, *CRP* levels and injuries).[39]

A holistic perspective of each individual player allows the practitioner to gain context into the cause of disturbance in the players physiological state and can therefore inform the subsequent course of action. For example, from a clinical perspective, the *POC* measurement of *CRP* alongside the self-reporting of illness symptoms could be used as an indicator of infection and identify players that require medical review. Further assessment by medical staff may include collection of body temperature to assess if URS are due to infectious or non-infectious causes.[40] On the other hand, when biomarker 'red flags' of physiological relevance are detected (e.g. *CRP* outside *IARR*), recovery strategies may be targeted. The dichotomy between the effects of acute and chronic exposure to inflammatory stimuli (i.e. the concept of hormesis) is important to consider when prescribing recovery strategies.[3, 9, 36] For example, if during a period where recovery is the priority (as opposed to adaptation), and a player presents with an abnormal profile, acceleration of inflammation resolution via targeted recovery practices may represent a practical approach for preventing the development of excessive systemic inflammation and promote the timely recovery of the player. This may also help progress away from the chronic use or 'one-size fits all' approach to the use of anti-inflammatory interventions by identifying periods where an individual may require (or not require) intervention, resulting in a more targeted and potentially more effective intervention. Indeed, the monitoring of inflammation in conjunction with other recovery metrics may provide insights into the physiological stress a player is experiencing (i.e. mechanical or metabolic stress) so that recovery strategies that match the associated physiological stress can be targeted to further individualize and accelerate the recovery process.[41] However, it is important to account for the importance of player belief [42] and note that any potential gains from such strategies remain futile in the absence of getting the basic principles of recovery (i.e. nutrition, hydration, sleep hygiene and appropriate rest) in place.[36]

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Practical Applications

1. There is a need to establish baseline values whilst the player is in a rested and healthy physiological state (e.g. frequent *POC* testing could be performed when the player returns from the off-season period) in order to gather baseline values and help establish *IARR* for each player prior to the commencement of the season.
2. The practitioner should look to apply *IARR* for each testing day (i.e. MD+1, MD+2 etc), instead of only detecting deviations from the recovered state or fatigued state (i.e. pre or post-match).
3. A practical approach is recommended whereby practitioners intuitively observe trends in the data and judge abnormal data points identified by *IARR* in a live manner (i.e. at the time of testing) based on a combination of factors including statistical methods, their own domain knowledge and relevant context collected from the player, staff and other data streams, so that meaningful changes can be distinguished from natural variations in the biomarker.
4. Merging biomarker data with other data sources routinely collected in the high performance environment and communication amongst key stakeholders including the sports scientist, performance nutritionist, team doctor and the player themselves is vital in understanding the biomarker results and informing subsequent course of action to protect player health and promote recovery and performance.

This is the first study, to our knowledge, to explore the application of *IARR* on longitudinal blood biomarker data in professional soccer and the *EPL*. Inherent limitations exist as a consequence of collecting data in professional soccer players. Moreover, given that training status is a factor dictating the magnitude of the inflammatory response, the applicability of our findings may be limited to players competing at a professional level. However, the nature of this sample, i.e. highly-trained EPL players in a 'real-world' setting strengthens the ecological validity and novelty of our data. Also, future work is required to more fully understand meaningful changes in inflammatory biomarkers in longitudinal biomarker tracking (i.e. the robustness of the relationship between changes in *CRP* and relevant negative outcome indicators). Anecdotally, the question practitioners often ask when encountering inflammation data such as those presented in the Player 4 case description is – *is this normal for him?* There is clear rationale as to why prolonged states of inflammation places the soccer player at risk of under-recovery and that players competing whilst experiencing the symptoms associated with low-grade inflammation may perform below their potential or even have an increased risk of injury.[36] However, to the authors knowledge, there are no data examining the degree to which a change in *CRP* promotes subsequent changes in negative outcome indicators such as decrements in physical or technical performance, illness, and injury. If match performance data demonstrates that players are coping physically, despite elevated levels of inflammation, then practitioners can legitimately question the relevance of such data.[16] Indeed, a greater understanding of the functional relevance of *CRP* in the real world applied environment may assist the practitioner in validating the decision to adjust a players training load in order to resolve an increase in *CRP* in the interval between successive matches. Moreover, with a longer-term perspective, chronic inflammation is a unifying feature of several disease states. Whilst these do not seem relevant to young healthy players, there is increasing attention on the long-term health of athletes as some sports are known to be strongly associated with disease continuums either during or post career.[43] Inflammation has been described in the pathogenesis of heart disease in American Football players[44] and is a key mediator in the long-term recovery from traumatic brain injury[34] – a potential mechanism for lifelong depression.[45] This may be of particular relevance for brain health in

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combat sports, which are often characterized by high concussion rates. [46] This in itself provides rationale for further research to understand healthy vs. damaging levels of inflammation in elite athletes and long-term studies are warranted to define the relative contribution of inflammation in the pathogenesis of disease in athletes and the effects of repeated inflammatory episodes on career longevity and on post-career health.

Conclusion

In conclusion, we present an individualized approach to biomarker monitoring which could help guide the practitioner's interpretation of individual biomarker data. *IARR* serve more as an early warning system than a diagnostic tool. Thus, significant care is needed to prevent misuse and misinterpretation when implementing this strategy in real-world settings. Future work is required to more fully understand meaningful changes in inflammatory biomarkers in longitudinal biomarker tracking and transform this promising strategy into an efficient monitoring tool in professional soccer.

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Declaration of Interest
The authors declare the following conflicts of interest relevant to the content of this study. Nathan A. Lewis, Georgie Bruinvels, Andrew Barr and Charles R. Pedlar are employees or consultants with Orreco Ltd. John Newell is the Principal Investigator of the Orreco Ltd-funded research project in the Insight Centre for Data Analytics, University of Galway. Orreco Ltd provide blood biomarker services to elite athletes.

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Appendices

Appendix 1: Figures

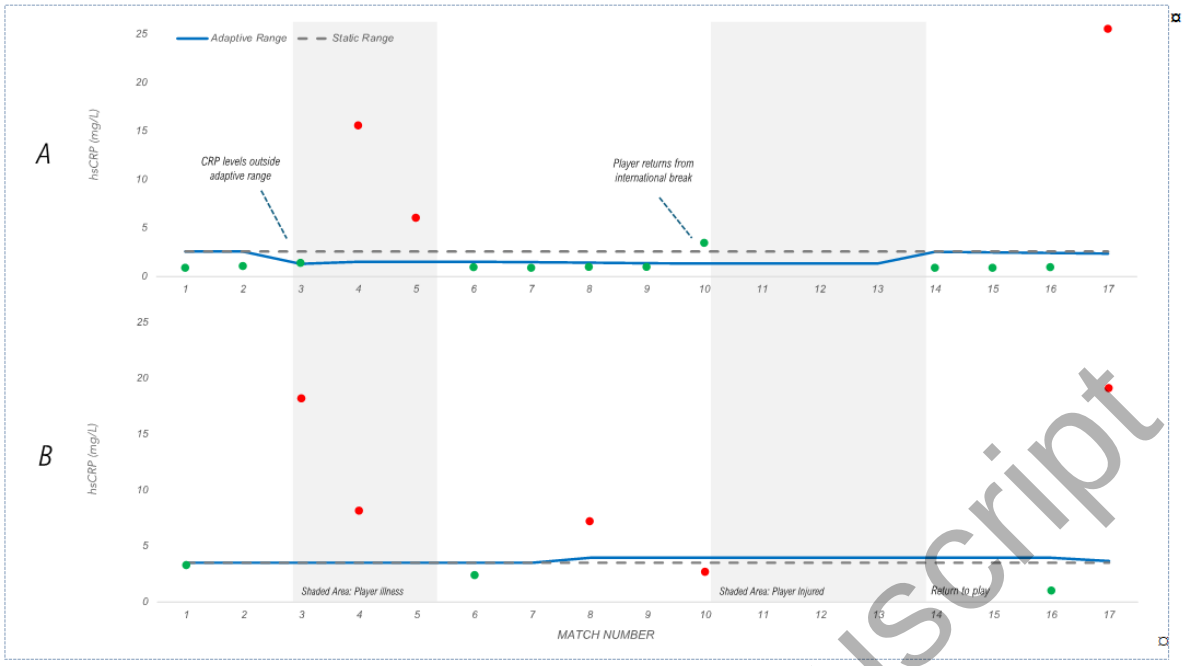

Figure 1: Player 1: A series of longitudinal biomarker values of (A) Pre-Match Measures (B) Post-Match Measures with static reference ranges (dashed line) and dynamic reference ranges using Approximate Expectation-Maximisation (solid blue line). Post-match measurements were taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) as follows; MD+1 (Matches 1, 3, 11, 14, 15), MD+2 (Matches 6, 8, 10, 12, 16), MD+3 (Match 7). Data points excluded from the computation of the reference ranges are highlighted in red. *hsCRP (high-sensitivity C-Reactive Protein).*

1

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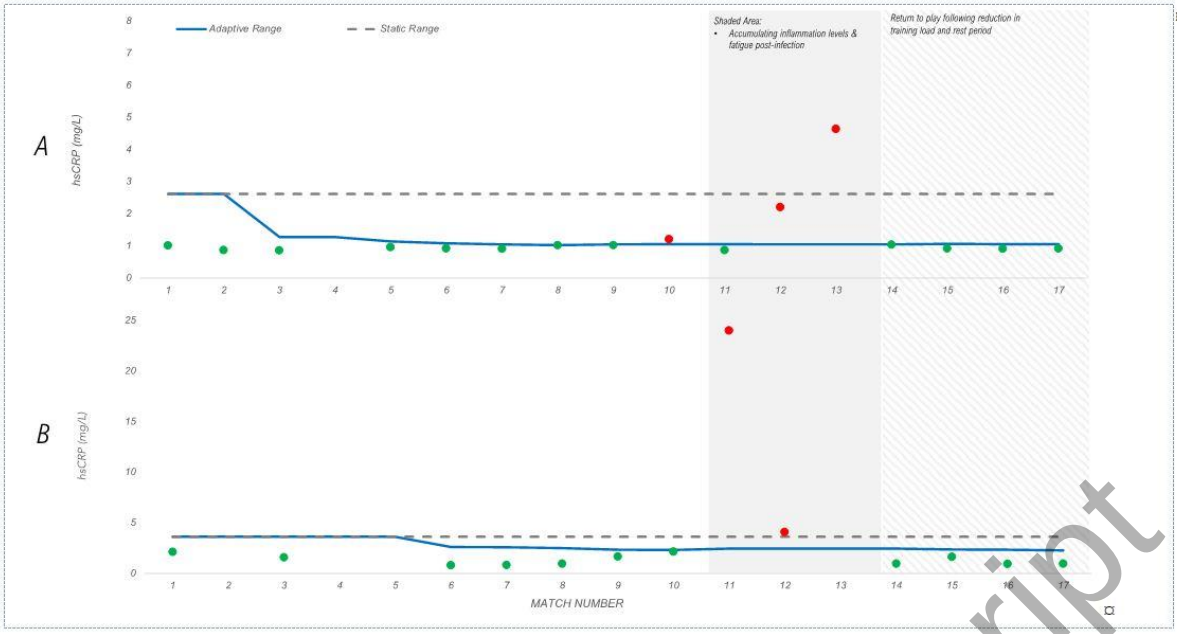

Figure 2: Player 2: A series of longitudinal biomarker values of (A) Pre-Match Measures (B) Post-Match Measures with static reference ranges (dashed line) and dynamic reference ranges using Approximate Expectation-Maximisation (solid blue line). Post-match measurements were taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) as follows; MD+1 (Matches 1, 3, 11, 14, 15), MD+2 (Matches 6, 8, 9, 10, 12, 16, 17), MD+3 (Match 7). Data points excluded from the computation of the reference ranges are highlighted in red. *hsCRP (high-sensitivity C-Reactive Protein).*

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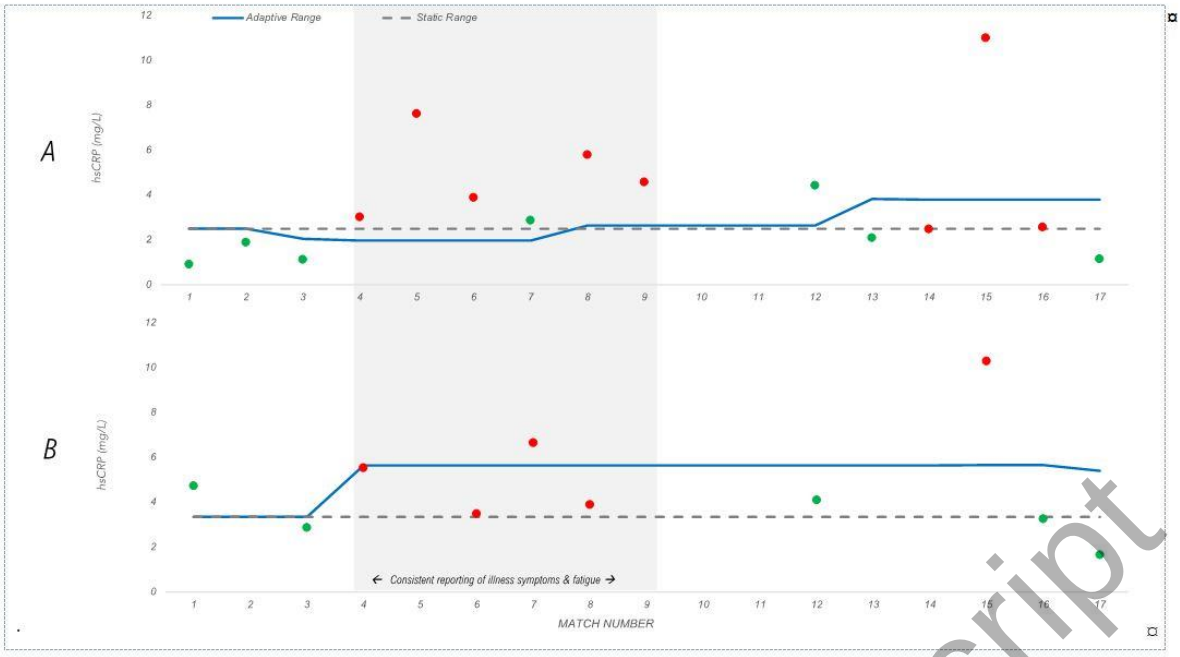

Figure 3: Player 3: A series of longitudinal biomarker values of (A) Pre-Match Measures (B) Post-Match Measures with static reference ranges (dashed line) and dynamic reference ranges using Approximate Expectation-Maximisation (solid blue line). Post-match measurements were taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) as follows; MD+1 (Matches 1, 3, 4, 15), MD+2 (Matches 6, 8, 12, 16, 17), MD+3 (Match 7). Data points excluded from the computation of the reference ranges are highlighted in red. *hsCRP (high-sensitivity C-Reactive Protein).*

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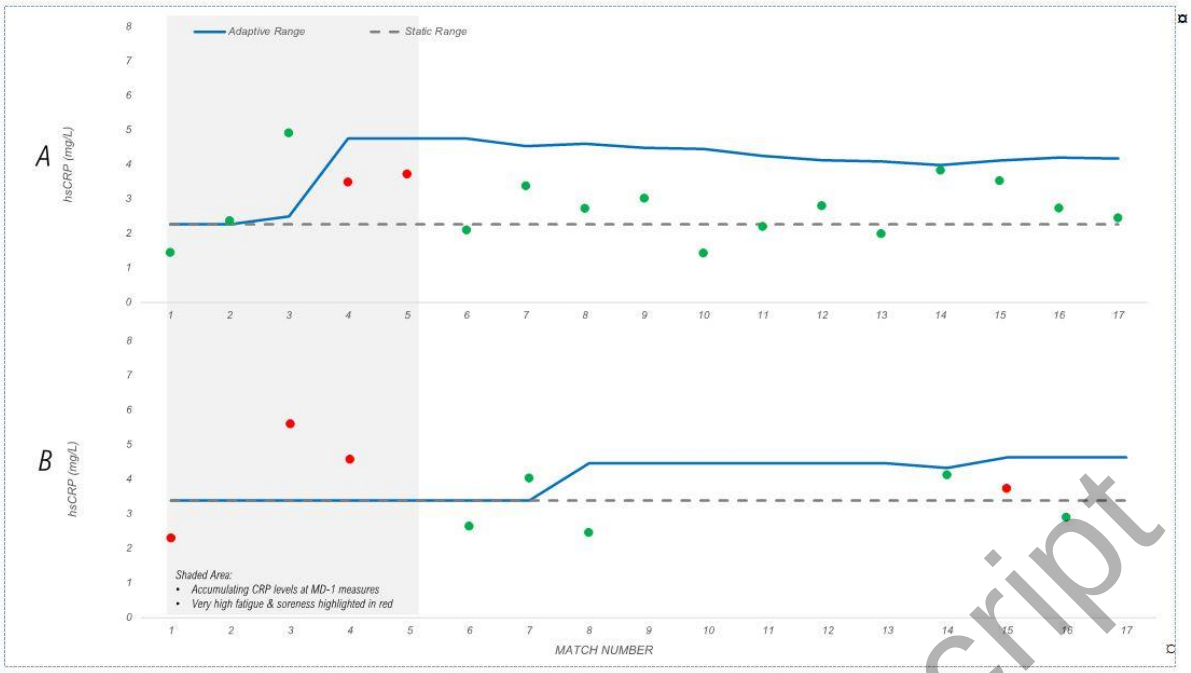

Figure 4: Player 4: A series of longitudinal biomarker values of (A) Pre-Match Measures (B) Post-Match Measures with static reference ranges (dashed line) and dynamic reference ranges using Approximate Expectation-Maximisation (solid blue line). Post-match measurements were taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) as follows; MD+1 (Matches 1, 3, 4, 14, 15), MD+2 (Matches 6, 8, 16), MD+3 (Match 7). Data points excluded from the computation of the reference ranges are highlighted in red. *hsCRP (high-sensitivity C-Reactive Protein).*

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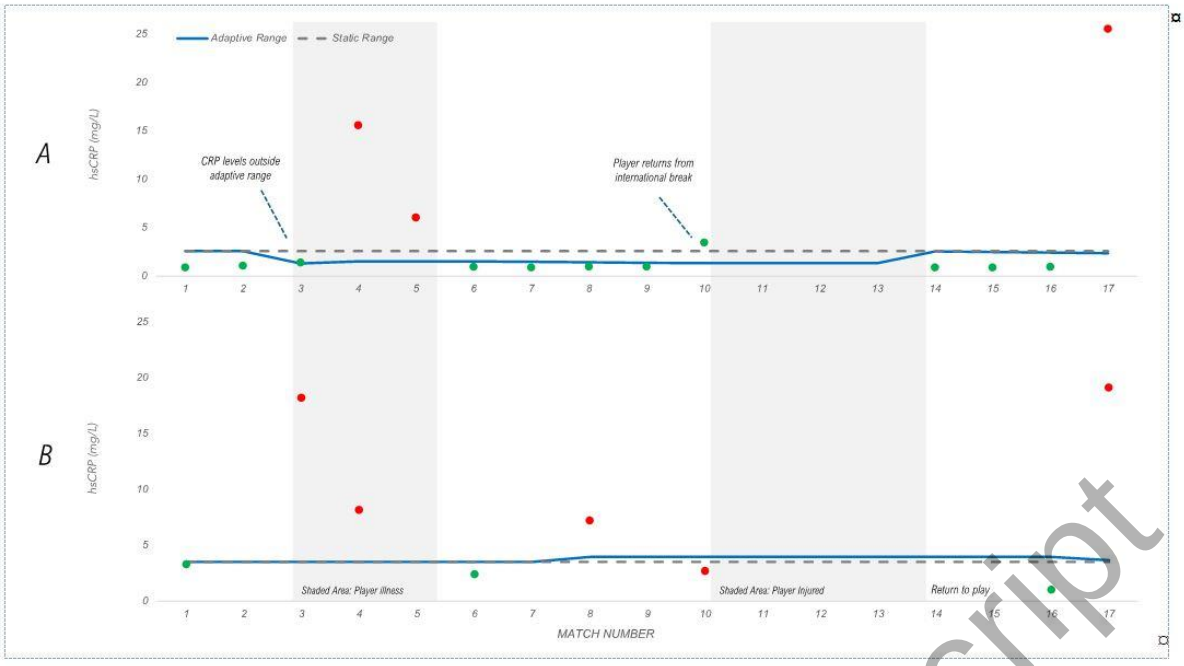

Figure 5: Player 5: A series of longitudinal biomarker values of (A) Pre-Match Measures (B) Post-Match Measures with static reference ranges (dashed line) and dynamic reference ranges using Approximate Expectation-Maximisation (solid blue line). Post-match measurements were taken on either match day +1 (MD+1), match day +2 (MD+2), match day +3 (MD+3) as follows; MD+1 (Matches 1, 3, 4, 14, 15), MD+2 (Matches 6, 8, 10, 16, 17). Data points excluded from the computation of the reference ranges are highlighted in red. *hsCRP (high-sensitivity C-Reactive Protein).*

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Appendix 2: Tables

Table 1: The overall performance of the two static and adaptive reference ranges in detecting abnormal observations.

CRP	Method	Sensitivity	Specificity	# of subjects
Post match CRP	**IARR**	0.852	0.868	18
	Static RR	0.815	0.904	18
Pre match CRP	**IARR**	0.878	0.891	22
	Static RR	0.862	0.910	22

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Table 2: Performance comparison of two methods based on the ratio of intra-individual to inter-individual variability

Ratio	CRP	Method	Sensitivity	Specificity	# of subjects
<0.25	**Post match CRP**	IARR	1.000	0.923	4
		Static RR	0.500	1.000	4
	Pre match CRP	IARR	1.000	0.949	11
		Static RR	0.833	1.000	11
[0.25,0.5)	**Post match CRP**	IARR	1.000	0.905	3
		Static RR	0.875	1.000	3
	Pre match CRP	IARR	-	-	0
		Static RR	-	-	0
[0.5,1)	**Post match CRP**	IARR	0.792	0.871	6
		Static RR	0.854	0.833	6
	Pre match CRP	IARR	1.000	0.897	2
		Static RR	0.500	0.933	2
[1,2)	**Post match CRP**	IARR	-	0.697	2
		Static RR	-	0.938	2
	Pre match CRP	IARR	-	0.861	3
		Static RR	-	0.931	3
[2,5)	**Post match CRP**	IARR	0.733	0.867	3
		Static RR	0.933	0.800	3
	Pre match CRP	IARR	0.667	0.816	5
		Static RR	1.000	0.728	5
>5	**Post match CRP**	IARR	-	-	0
		Static RR	-	-	0
	Pre match CRP	IARR	0.750	0.714	1
		Static RR	0.875	0.714	1

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Table 3: Player 1 Case Description

Player 1 - Case Description
Performance comparison of methods: ▪ Data presented for Player 1 highlights a scenario whereby *IARR* is the most effective method for monitoring *CRP* levels, as the intra-individual variability is considerably lower than inter-individual variability. As can be seen in **Figure 1A** and **Figure 1B**, both methods achieved 100% sensitivity for pre-match *CRP* measurements, however, the *IARR* demonstrates superior sensitivity (i.e. 100%), compared to the static reference range's 75% at post-match. The heightened sensitivity in *IARR* comes with slightly lower specificity, for example the *CRP* false positive identified by the *IARR* prior to match 8. Indeed, when the intra-variability is low, the *IARR* is highly sensitive to minor changes in biomarker levels because of its narrower range. *Case Description:* ▪ Player 1 experienced an increase in high-intensity workload during matches 11-14, with THSR ranging from 0.7-0.97 km across the 4 games measured, the highest experienced by the player up to that point (player average THSR = 0.68km). Increases in THSR have been shown to be associated with augmented sensations of fatigue in elite soccer players,[29] and the player persistently reported feeling very fatigued during this period. Testing administered after a match may be valuable when resting values of biomarkers might not reveal any concerns, but biomarker responses to an acute stress would more sensitively detect recovery issues and heavily fatigued states,[30] as was the case for Player 1. Notably, the post-match measure after match 12 was outside the players *IARR* but within the squads static reference range of 3.59 mg/L (**Figure 1B**). Susceptibility to illness and infection is exacerbated during periods of heavy competition,[31] and the increase in workload during this period may have contributed to the upper respiratory tract infection (URTI) observed post-match 14 and pre-match 15, in the upper right quadrants of **Figure 1A** and **Figure 1B**. Moreover, post-match measures 14 and 15 were outside both the *IARR* and the static range but within the clinical range of <5.1 mg/L for soccer players previously defined by Meyer et al.,[27] and therefore the static and *IARR* correctly identified atypical measures which may be of relevance to the sports science and medical team. The MD-1 measure of *CRP* also proved valuable in this case, with the athlete considerably outside the upper reference limit of his *IARR*, but notably within the clinical range.[27]

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Table 4: Player 2 Case Description

Player 2 - Case Description
Performance comparison of methods: ▪ Player 2 generally presented with low intra-individual variability for *CRP* pre-match (**Figure 2A**) and post-match (**Figure 2B**), resulting in a narrow *IARR* for *CRP* and as a result more sensitive to slight changes. The performance of the static reference range for post-match *CRP* were comparable to the *IARR* (both achieving 100% sensitivity and specificity), however, the *IARR* for pre-match *CRP* demonstrates superior sensitivity, achieving 100% compared to the static reference range's 33%, indicating the static reference range identified only 1 of the 3 abnormal values correctly. *Case Description:* ▪ Abnormal values were detected by both methods post-game 11, with the athlete presenting with *CRP* levels of 24 mg/L, due to an URTI (**Figure 2A**). It is plausible that the recovery period post-match led to a short-term suppression of the immune system, giving pathogens a chance to gain a foothold, and leading to a greater susceptibility to infection, as described in the "open window theory".[31] Furthermore, we observed a trend for cumulative post-match *CRP* levels prior to the infective episode (**Figure 2B**), from match 7-9 (119% relative change). It is noteworthy that this rise in *CRP* levels prior to infection was within the players IARR, which emphasises the importance of human interpretation in assessing a gradual change of concern. Moreover, the athlete reported upper respiratory symptoms (URS) a week prior to the infective episode that were outside his *IARR* but notably within the static reference range (**Figure 2A**). Accumulating fatigue may persist post-infective episode, particularly in the athlete who continues to compete,[32] as was the case here, evidenced by the augmented levels of *CRP* and the consistent reporting of low energy levels in the week that followed (**Figure 2A**). Notably, this abnormality was detected by the *IARR*, but not the static reference range, for this athlete pre-match 11. Systemic inflammation was subsequently resolved after an enforced reduction in training load after match 12.

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Table 5: Player 3 Case Description

Player 3 - Case Description
Performance comparison of methods: ▪ Player 3 presented with high intra-individual variability for *CRP* pre-match (**Figure 3A**) and post-match (**Figure 3B**), resulting in a wide *IARR* for *CRP*. In this case, the static reference range demonstrates superior sensitivity at pre-match (88%) and post-match measures (100%), compared to the *IARR* 75% and 40% respectively. This is an example where *IARR* produces reference ranges that are wider than the static reference ranges, which enhances specificity but at the expense of sensitivity. In this case, the *IARR* is less effective at identifying outliers because of its wider range, although it produces less false positives that the static reference range. It should also be noted that the number of healthy records for this player is quite low, while, the *IARR* requires more data points to be fully adapted and personalized. For example, in the post-match data, there are only 2 healthy records before a series of 4 consecutive abnormal values. Similarly, in the pre-match data, there are only 3 healthy records before the occurrence of abnormal observations. *Case Description:* ▪ We present evidence suggestive of impaired inflammatory regulation in an illness-prone athlete in **Figure 3A** and **Figure 3B**. The athlete reported illness symptoms or was classified as ill by the medical staff on 8 occasions out of the 15 measures taken at MD-1 (**Figure 3A**) and 5 out of the 10 post-match measures and consistently reported low energy levels at these time points (**Figure 3B**). These measures were excluded from the calculation of the athletes reference ranges, however, the high intra-variability in his biomarker responses that were included throughout the testing period resulted in a wide *IARR* for this player. Notably, the player reports illness symptoms post-match 4, 6 and 8, with *CRP* levels outside the static range but notably within his *IARR* and identified as typical or normal for that individual. In this scenario, the *IARR* therefore fails to detect true abnormalities and does not identify data points of clinical relevance. This, however, is mainly due to the lack of sufficient data prior to post-match 4, which is necessary for the *IARR* to be fully adapted and more personalized. Disruptions in the balance between pro- and anti-inflammatory cytokines can lead to a loss of inflammatory control in illness prone athletes[33] and could explain the high variability in responses we observed. Cox et al. [33] presented evidence of lower resting IL-8, IL-1ra, and IL-10 concentrations and a more potent IL-6 response (IL-6 acts on hepatocytes to stimulate the production *CRP*) to treadmill running in illness-prone compared with healthy athletes. Indeed, it is noteworthy, that the magnitude of the inflammatory responses for this player were considerably higher than the rest of the squad (**Figure 3B**). Additionally, the lower IL-8, known as neutrophil chemotactic factor, would suggest a lesser propensity for neutrophil recruitment in the illness prone athlete, which could be dis-advantageous in instances of infection, and explain why resistance to infection was reduced.

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Table 6: Player 4 Case Description

Player 4 - Case Description
Performance comparison of methods: **Figure 4A** and **Figure 4B** visualises data from a player with relatively higher *CRP* results compared to the rest of the players in the sample and as a result the majority of his test results were outside the defined static reference ranges. This is an example where, although the static reference range identifies most values as abnormal, the *IARR* learns from the individual's trajectory and adjusts the reference ranges based on the player's variability in *CRP*, thereby producing a lower false positive rate compared to the static reference ranges (13% vs. 67% in pre-match data). *Case Description:* ▪ The static reference range identified most values as abnormal, which is clearly not the case for this player. However, whilst the chronic state of inflammation may be considered normal for this player, it is important to make the distinction between what is normal and what is healthy, given the link between chronic inflammation and a host of recovery issues[9] and sport specific medical problems.[34] The elevated levels of *CRP* captured at MD-1 of match 3, would suggest the player competed whilst not entirely recovered, and it is noteworthy that the player reported feeling very sore and fatigued at subsequent pre- and post-match measures. The consistent exposure to competition that followed, led to further development of a chronic state of low-grade inflammation. This data point, identified as a false positive by the *IARR*, may have served as an early warning system at the time of testing, however, the lack of contextual information to substantiate this, resulted in the inclusion of this data point in the calculation of the *IARR* and the widening of the players *IARR*. The *IARR* will learn from previous measurements, and the subsequent measures were considered typical for this player. Nevertheless, the *IARR* were still trying to adapt whenever a new observation was collected for the individual. Notably, with the exception of the data points highlighted in red, the player consistently reported the same subjective wellness throughout the study period. A prolonged state of low-grade inflammation has been described to predispose susceptible individuals to the development of chronic overuse tendinopathies,[35] and Player 4 suffered a tendon injury in January. Although speculative, it is plausible that the demands of the congested *EPL* winter schedule that followed our study period, may have further exacerbated the athletes' chronic state of inflammation leading to a greater susceptibility to injury.

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Table 7: Player 5 Case Description

Player 5 - Case Description
Performance comparison of methods: ▪ In this case, both methods achieved 100% sensitivity for pre-match *CRP* measurements, correctly identifying points of clinical relevance (due to illness) at matches 4, 5 and 17. Both methods achieved 80% sensitivity for post-match *CRP* measurements, as the injury detected at match 10 was within the *IARR* and static reference ranges. In this example, the *IARR* comes with slightly lower specificity for pre-match *CRP*, as a result of the false positives identified by the *IARR* prior to match 3 and 10. *Case Description:* ▪ Data from Player 5 suggests that *IARR* serves as an effective early warning system for identifying players at risk of illness and injury. Notably, the pre-match *CRP* measure at match 3, in the lower left quadrant of **Figure 5A**, were outside the athletes *IARR*, but within the static range for the squad at MD-1. This is an example where a pre-match CRP false positive were identified by the *IARR* prior to an actual abnormal value (due to an illness) post-match 3 (**Figure 5B**). Systemic inflammation remained elevated as the player continued to compete, and he consistently reported low energy levels alongside illness symptoms during this period. Another pre-match *CRP* false positive (i.e. an early warning) was identified by the *IARR* prior to an actual abnormal value post-match 10 (**Figure 5B**). The player returned from an international break prior to the measure taken at MD-1. The abnormal value detected pre-match 10, was outside both the *IARR* and static reference ranges, suggesting the player may not have been fully recovered. In addition to the demanding fixture schedule associated with international duty, inducing muscle fatigue and soreness, the travel requirements and potential disruptions to the habitual sleep patterns of the player,[23] may have affected his recovery status and increased background inflammation levels prior to match 10. It is noteworthy, that the player subsequently suffered an injury during match 10 and missed the next 3 matches of the season. Exposure to competition whilst players are still 'recovering and regenerating' and experiencing the symptoms associated with muscle damage and inflammation, may increase the risk of injury.[36] Although speculative, it is plausible that the abnormal inflammation levels at MD-1, detected by both the static and *IARR*, indicated that the athlete was under-recovered and at a greater risk of injury.

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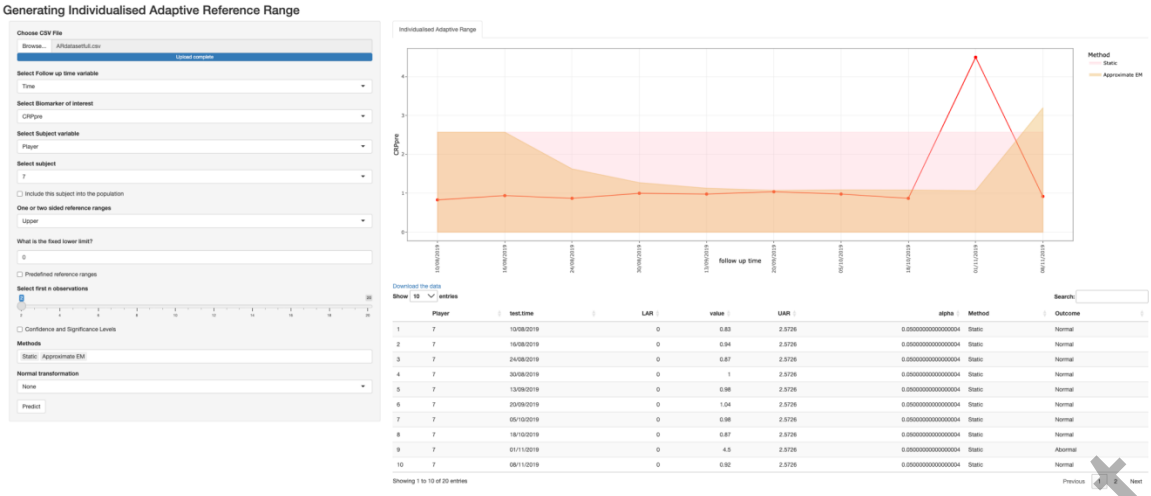
Appendix 3:



Appendix 3: A screenshot of the developed interactive web-based application for generating static and dynamic reference ranges. This example using an anonymised dataset (written in long format) showcases the use of the Shiny application (https://pci-nuig.shinyapps.io/barshinydemo/) as a guideline for other users, demonstrates the practical implementation of the methodology and highlights its application to real-world data.

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